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MicroRNA bioinformatics in precision oncology: an integrated pipeline from NGS to AI-based target discovery

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Abstract

Despite the growing recognition of microRNAs (miRNAs) as critical biomarkers in cancer, current approaches to their analysis remain fragmented, disjointed, and poorly integrated with emerging computational advances. This lack of cohesion limits progress toward reproducible and clinically actionable biomarker discovery. To address this unmet need, we present a review that unifies the latest findings and tools in bioinformatics, machine learning (ML), and large language models (LLMs) for miRNA analysis in oncology, thereby bridging a significant methodological gap in the field. We begin by critically synthesizing, benchmarking, and evaluating algorithms, including miRDeep2 and DIANA-miRPath, within a functional pipeline that spans next-generation sequencing (NGS) data processing to multi-omics integration. Building on this foundation, we review ML-augmented layers incorporating supervised and deep learning (DL) algorithms, specifically support vector machines (SVMs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs), to enable robust miRNA signature identification, classification, and target prediction. Furthermore, we explore the integration of generative models and LLMs to support hypothesis generation and enhance reproducibility in biomarker discovery workflows. This comprehensive framework enhanced with artificial intelligence (AI) is contextualized through cancer-specific datasets, with particular emphasis on translational applications for early detection, prognosis, and therapy selection. By systematically organizing fragmented methodologies into a scalable and reproducible pipeline, our work provides a strategic roadmap to accelerate the development of miRNA-based precision cancer.

Keywords MicroRNA biomarkers · Machine learning in oncology · AI-assisted biomarker discovery · Multi-omics integration · Next-generation sequencing · Precision oncology

Introduction

MicroRNAs are a class of small noncoding RNA molecules that play a critical role in the regulation of gene expression (Matulić et al. 2022). They bind to complementary

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sequences in the 3'untranslated regions or open reading frames of target messenger RNAs (mRNAs), leading to mRNA degradation or reduced protein production (Wu et al. 2016). Their involvement in cancer biology is well established, as they can function as either oncogenes or tumor suppressors (Dey et al. 2024; Han et al. 2025). The differential expression of miRNA is often associated with tumour development and progression. The advantages of miRNAs as biomarkers lie in their stability in biofluids, such as blood, urine, and saliva, making them ideal candidates for non-invasive biomarkers (Mestry et al. 2023). Unlike mRNAs, which are prone to rapid degradation, miRNAs exhibit resistance to enzymatic breakdown, enabling their detection and quantification in clinical samples with high reliability (Glinge et al. 2017; Gustafson et al. 2016).

In addition, miRNAs display tissue and tumor type specificity, which means that different expression patterns can be used to classify different types of cancer and even distinguish between subtypes (Heneghan et al. 2010; Lan et al. 2015). Beyond diagnosis, miRNAs have significant prognostic and predictive value in oncology. Specific miRNAs, such as miR-21, are frequently overexpressed in multiple tumors and have been associated with a poor prognosis, aggressive tumor behaviour, and resistance to therapy (Pan et al. 2010). Such findings underscore the clinical potential of miRNAs to guide personalized treatment decisions. As research progresses, the integration of miRNA-based biomarkers into routine clinical practice could revolutionize early cancer detection, risk stratification, and therapeutic monitoring, ultimately improving patient outcomes.

Bioinformatics plays an increasingly crucial role in advancing our understanding of microRNAs and their implications in cancer biology. By combining powerful computational tools with high-throughput sequencing technologies, researchers can now analyze vast amounts of miRNA expression data from various biological sources to identify distinct miRNA signatures associated with different types and stages of cancer. This integration of bioinformatics has enabled researchers not only to detect miRNAs with high sensitivity and precision, but also to predict their potential regulatory targets and biological functions, shedding light on their role in cancer initiation, progression, and metastasis (Qian et al. 2005; Qian et al. 2003; Wolde et al. 2025).

Bioinformatic pipelines can now identify differentially expressed miRNAs in various cancers, facilitating the discovery of novel biomarkers for early detection, diagnosis, and prognosis (Hu et al. 2018; Lin et al. 2012). Furthermore, the use of databases like TCGA (The Cancer Genome Atlas), GEO (Gene Expression Omnibus), and miRBase has allowed researchers to validate the findings in multiple studies, enhancing the reproducibility and generalizability of miRNA biomarkers (Tomczak et al. 2015; Barrett et al. 2013; Kozomara and Griffiths-Jones 2011). Furthermore, computational tools such as DIANA-miRPath, TargetScan, and KEGG-based pathway analysis provide deeper insights into how miRNAs regulate complex signalling networks involved in tumorigenesis, helping to identify potential therapeutic targets (Tastsoglou et al. 2023a; Agarwal et al. 2015; Rafiepoor et al. 2025).

Unlike siRNA therapies, which have already reached clinical use with drugs such as patisiran approved for hereditary transthyretin-mediated amyloidosis, miRNA mimics and inhibitors are still confined to preclinical studies or early clinical trials (patsnap, S.b. 2025; Ju 2025). As we move toward personalized medicine, bioinformatics will continue to revolutionize miRNA research by enabling the integration of miRNA data with other omics layers, such as transcriptomics, proteomics, and genomics. Multi-omics approaches will allow for an even more comprehensive understanding of the regulatory mechanisms in cancer and facilitate the development of miRNA-based diagnostic and therapeutic strategies. In addition, the use of machine learning (ML) and artificial intelligence (AI) in bioinformatics will further enhance the identification of miRNA biomarkers with high predictive power, accelerating the transition of miRNA-based applications into clinical practice.

In the near future, the incorporation of miRNA profiling into routine clinical settings, combined with bioinformatics-driven analysis, has the potential to transform cancer management. Noninvasive miRNA-based tests could enable early cancer detection, better stratification of patients according to risk, and more precise monitoring of treatment responses. Ultimately, as bioinformatics continues to advance, it will shape the future of cancer research and clinical practice, unlocking new avenues for personalized therapies and improving patient outcomes. This paper will focus on the bioinformatics analysis pathways and will also shed light on the potential use of large language models (LLMs) in microRNA data analysis in cancer.

Brief outline of data sources and analysis pipeline

The study of microRNAs in cancer uses data from a combination of clinical and cell culture samples processed through NGS, publicly available databases, and clinically annotated patient records. Clinical samples, such as tumor biopsies, blood, and other biofluids, undergo NGS-based profiling to identify miRNAs and quantify their expression levels. Additionally, cell culture models, such as cancer cell lines or patient-derived organoids, provide controlled environments to validate the roles of specific miRNAs. Open source databases such as TCGA, GEO, miRBase, and MirGeneDB offer curated datasets and annotations, allowing researchers to perform cross-study validation and further analysis (Tastsoglou et al. 2023a; Agarwal et al. 2015). Clinical records, which contain information on tumor grade, stage, and patient survival, allow researchers to correlate miRNA expression patterns with disease progression and response to treatment.

A typical NGS data processing pipeline follows a stepwise approach starting with quality control (QC), which identifies errors or biases in raw sequencing data, such as adapter contamination or low-quality reads. The next step, adapter trimming, removes these unwanted sequences (adapters) that were added during the sequencing process but do not represent actual miRNAs (Dijk et al. 2014). The read alignment follows, where the sequencing reads are mapped to a reference genome or a miRNA-specific database (e.g., miRBase) to identify and quantify the miRNA sequences present. Alignment ensures that miRNA sequences are accurately mapped to correct genomic locations, allowing for precise expression quantification. After alignment, normalization is applied to adjust for sequencing depth and technical biases, ensuring that comparisons of miRNA expression levels across different samples are meaningful. Common methods of normalization include reads per million (RPM), which adjusts for differences in sequencing depth, and trimmed mean M-values (TMM), which corrects for compositional differences between samples (Li et al. 2020).

Once data has been preprocessed, differential expression analysis identifies miRNAs that are significantly upregulated or downregulated between cancer and normal samples, using statistical tools such as DESeq2 or edgeR (Niedziela et al. 2022). To gain insight into the biological roles of differentially expressed miRNAs, functional annotation and pathway enrichment tools (e.g., DIANA-miRPath, miRNet) are used to link these miRNAs to potential target genes and signaling pathways involved in cancer (Papadopoulos et al. 2009; Robinson et al. 2010). Integrating these results with data from other omics layers, such as transcriptomics and proteomics, provides a broader understanding of miRNA-mediated regulatory networks. Figure 1 broadly outlines the steps involved in obtaining and processing NGS data. Every tool described here has been elaborated upon in greater detail in the later sections (Fig. 1).

Although the framework for miRNA data analysis is standard, variations in the pipeline can arise based on factors such as the sequencing platform used (e.g., Illumina, Ion Torrent, Oxford Nanopore) and the specific goals of the study. The choice of normalization methods, statistical models, and bioinformatics tools can all influence the results. Despite these variations, the overall objective of miRNA profiling remains consistent: to identify key biomarkers and pathways that can improve the understanding of cancer biology and help develop personalized therapeutic strategies.

Databases for microRNA analysis in cancer

Several databases have been developed to facilitate microRNA research, each offering unique functionalities for sequence annotation, expression profiling, target prediction, and pathway analysis. These databases vary in scope, with some focusing on experimentally validated data and others relying on computational predictions. Selecting the appropriate database is crucial to ensure reliable and meaningful insights in cancer-related miRNA studies. Among these resources, a major focus is on target prediction tools, which can be broadly grouped according to their underlying methodology. Sequence-based tools—such as TargetScan, miRanda, PITA, and PACCMIT-CDS—use base-pairing rules and evolutionary conservation to predict binding sites between miRNAs and mRNA targets (McGeary et al. 2019; Peterson et al. 2014; Šulc et al. 2015). Energy-based tools like PicTar, RNAhybrid, RNAduplex, and microT-CDS estimate the thermodynamic stability of miRNA:mRNA duplexes to identify potential regulatory interactions (Krek et al. 2005; Krüger and Rehmsmeier 2006; Tastsoglou et al. 2023b). Machine learning–based tools like MBSTar, MiRTDL, TarPmiR, and miRDB leverage datasets of experimentally validated interactions and sophisticated pattern-recognition algorithms to enhance specificity and sensitivity (Bandyopadhyay et al. 2015; Shuang et al. 2016; Chen and Wang 2020). Statistics-based methods, such as RNA22, utilize probabilistic patterns to highlight promising binding sites, while database-driven approaches (e.g., StarMirDB) mine curated experiment-derived datasets to annotate and rank miRNA targets (Loher and Rigoutsos 2012; Rennie et al. 2016). To help researchers navigate this complex landscape, Tools4miRs provides a comprehensive and actively maintained directory of over 170 functional miRNA databases and tools, covering sequence annotation, target prediction, expression analysis, and pathway enrichment. Importantly, it also indicates which resources are currently operational, enabling users to focus on active and reliable platforms for their analyses (Lukasik et al. 2016).

miRBase serves as the primary repository for known miRNA sequences, precursors, and genomic annotations. It is widely used for the identification and classification of miRNAs in various species. However, because miRBase includes both experimentally validated and predicted miRNAs, there is a possibility of false positives (Griffiths-Jones, et al. 2006; Ludwig et al. 2017). For example, one study claimed that fewer than 1881 human miRBase entries and about 16% of 7095 metazoan entries are supported by experimental data (Fromm et al. 2015; Fromm et al. 2022). In contrast, MirGeneDB3.0 provides a more stringent, manually curated collection of high-confidence

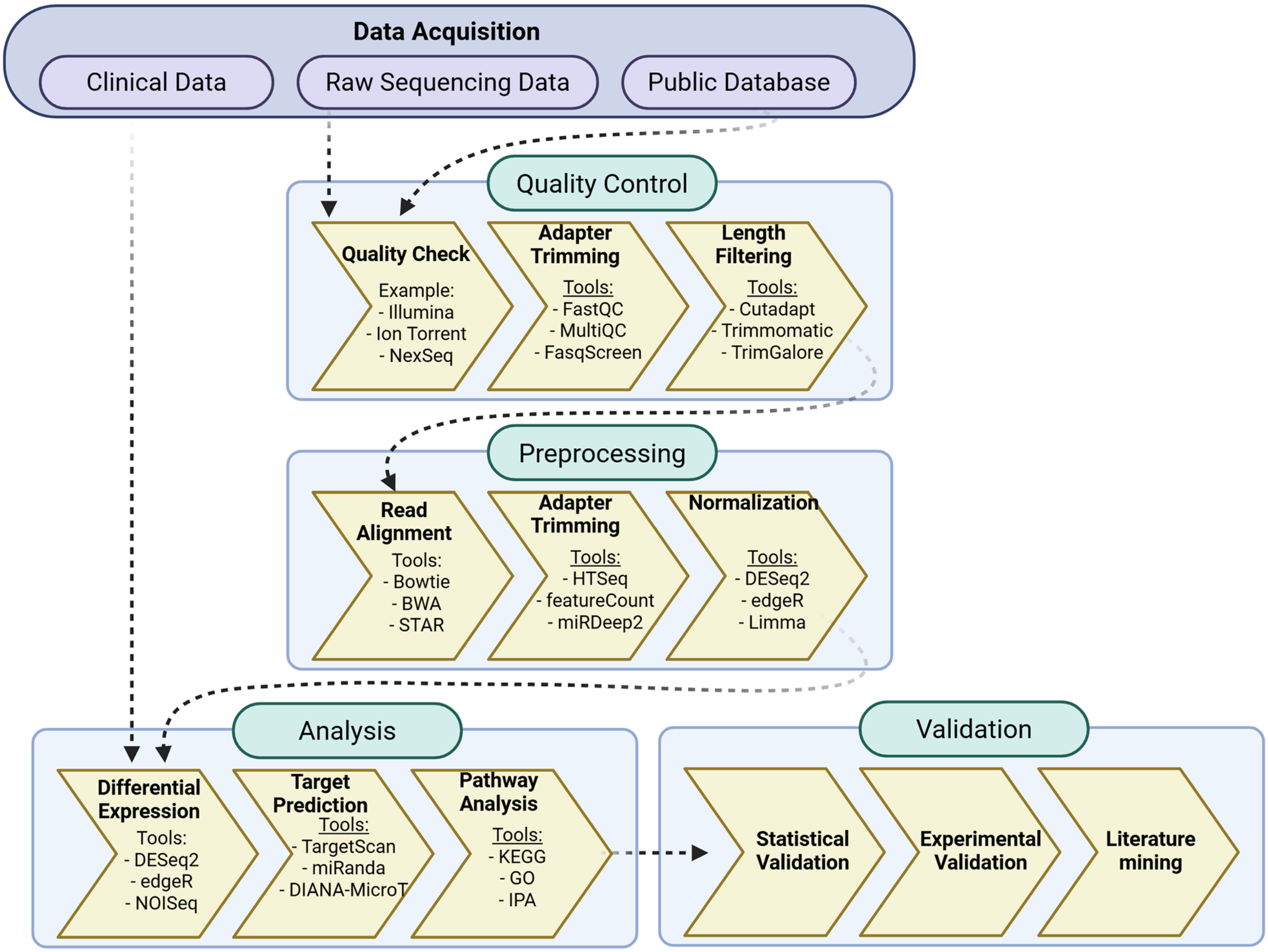

Fig. 1 A schematic representation of a typical bioinformatics data analysis pipeline. The process begins with collecting high-quality data from various sources, followed by a series of steps, including quality control and preprocessing. The refined data are then analyzed, visualized, and further integrated with other datasets to provide comprehensive insights (Created in BioRender. Dey, M. (2025a))

miRNAs, ensuring greater annotation accuracy. Unlike miRBase, MirGeneDB3.0 adopts a standardized nomenclature that makes orthologues across species readily identifiable, and it places particular emphasis on the careful curation of both mature and star sequences. Moreover, all entries are consistently remapped to the latest genome assemblies, further enhancing reliability and consistency in annotation (Fromm et al. 2015; Fromm et al. 2020; Clarke et al. 2025; Demirci and Allmer 2017).

In addition to foundational sequence repositories, several integrative databases and resources enhance the landscape of microRNA research. RNAcentral serves as a comprehensive platform that aggregates non-coding RNA sequences—including microRNAs—from over 40 specialized databases, offering unified access to both sequence and secondary structure information for a broad range of organisms (secondary structure integration, improved sequence search and new member databases. 2021). Another tool dedicated to the identification of miRNA gene promoters is DIANA-miRGen v4, providing a catalogue of experimentally validated and computationally predicted transcription start sites (TSSs) for human, mouse, and chicken miRNAs. Through integration with ENCODE and FANTOM5 datasets, miRGen v4 allows for detailed exploration of cell type–specific regulatory features, facilitating studies on transcriptional regulation and expression dynamics of miRNAs in physiological and disease contexts (Perdikopanis et al. 2021).

Recently, integrated prediction systems have emerged to address the limitations of single-method tools. miRgo is a novel platform freely available to the research community, combining the predictive power of 11 established algorithms with advanced feature selection (using mRMR) and a support vector machine (SVM) framework. Trained on experimentally validated targets from miRTarBase, miRgo achieves higher performance in accuracy and specificity compared to traditional approaches (Chu et al. 2020). While miRgo enhances prediction accuracy through algorithmic integration and machine learning, miRTargetLink 2.0 complements this by offering an interactive platform for biological interpretation. It provides a user-friendly interface to explore miRNA–target and pathway relationships, integrating validated and predicted associations from established databases (miRTarBase, mirDIP, miRDB, miRATBase), enriching them with pathway data (miRPathDB), and supporting dynamic network construction.

For studying miRNA expression in cancer, databases such as TCGA, GEO, and OncomiR provide valuable datasets. TCGA offers large-scale multi-omics data, including miRNA expression profiles linked to clinical parameters in different cancers, making it a critical resource for translational research (Liu et al. 2015; Wei et al. 2018). GEO, on the other hand, is a more general repository that hosts a wide range of high-throughput gene expression datasets, including miRNA sequencing (miRNA-seq) and microarray studies under various experimental conditions (NCBI 2025). OncomiR is specifically designed for cancer-related miRNA expression, offering survival analysis and prognostic information, making it particularly useful for identifying miRNA biomarkers associated with clinical outcomes (Wong et al. 2018).

The function is largely determined by their interactions with target genes, and various databases have been developed to predict and validate these interactions. TargetScan is one of the most widely used target prediction tools, leveraging evolutionary conservation and seed region matching to identify potential miRNA targets (McGeary et al. 2019). miRDB, in contrast, applies a machine learning-based algorithm trained on experimental data to improve prediction accuracy (Chen and Wang 2020). While these databases rely on computational models, TarBase provides a repository of experimentally validated miRNA-target interactions, offering a more reliable source for functional studies. Since computational predictions often yield false positives, integrating TarBase results with predictive tools enhances confidence in miRNA-target relationships (Skoufos et al. 2024; Karagkouni et al. 2018; Vergoulis et al. 2012).

A comprehensive analysis of microRNA data requires the integration of multiple databases, each offering distinct but complementary functionalities. Given the variability in prediction algorithms, data curation methodologies, and functional annotation frameworks across these platforms, a multidatabase approach is often required to ensure the robustness and reproducibility of miRNA-related findings in cancer research.

Pre-processing and data analysis

The analysis of microRNA data involves a series of computational and statistical techniques designed to accurately identify differentially expressed miRNAs and assess their biological relevance in cancer. The workflow typically includes QC, preprocessing, differential expression analysis, target prediction, pathway enrichment, and integrative multiomics analysis. Each of these steps plays a vital role in extracting meaningful insights from miRNA sequencing data and validating its significance in cancer biology.

Conversion of raw NGS files

For Illumina sequencing data, raw files are typically generated in BCL (Base Call) format, which contains base calls along with quality scores. Since Illumina is currently the most widely used second-generation sequencing platform, this step is common in most workflows. These BCL files must be converted into FASTQ format for downstream analysis. The conversion is usually performed using Illumina's BCL2FASTQ tool, which transforms BCL files into FASTQ files containing both sequence reads and their quality scores (Illumina. bcl2fastq. 2025). This step is crucial, as FASTQ files are the standard format required for most bioinformatics tools used in NGS data processing. BCL2FASTQ handles the conversion efficiently, ensuring the integrity of sequence reads and quality scores while also allowing for the demultiplexing of indexed reads from different samples.

Quality control

Raw sequencing reads undergo QC before analysis to identify any technical biases or artefacts. Tools such as FastQC are commonly used to assess the quality of sequencing reads, checking for issues such as adapter contamination (where adapter sequences from the sequencing process remain in the data), low-quality reads, and GC-content biases (Bioinformatics 2025). Once QC is completed, adapter trimming is performed using tools like Cutadapt or Trim Galore. Both tools serve a similar purpose, removing unwanted adapter sequences, but differ in some functionalities. Cutadapt is highly customizable, allowing users to specify multiple adapter sequences and apply various trimming strategies based on user-defined parameters. It can also handle multiple input formats and has options for quality-based trimming (Martin 2011). Trim Galore is a wrapper around Cutadapt and FastQC, making it more user-friendly and primarily focuses on trimming adapter sequences while integrating quality filtering steps. It is generally simpler and optimized for the trimming of high-quality data. After removal of the adapter, length filtering is applied to retain biologically relevant miRNA sequences, ensuring the exclusion of reads that fall outside this size range due to degradation or non-miRNA contaminants (Krueger 2015).

Read alignment an identification of microRNA

Identification of miRNAs from sequencing data relies on specialized tools that integrate read alignment with annotation, filtering, and structural evaluation. Among the tools for miRNA identification, miRDeep2 remains one of the most established. It relies on Bowtie2 for alignment and evaluates precursor sequences for canonical hairpin structures, mature/star strands, and Dicer processing signatures, combining these features with a probabilistic model to detect both known and novel miRNAs (Friedländer et al. 2012). Its strength lies in sensitivity, particularly for novel candidates, but benchmarking has shown a tendency toward false positives that require experimental validation (Mackowiak 2011). miRge2.0, by contrast, integrates alignment with a machine learning module for novel discovery, applying SVM models to features such as hairpin structure and isomiR (variant of canonical microRNA arising from variation in nucleotide addition, deletion, or substitution) composition. Its approach is more conservative than miRDeep2, reducing false positives and often outperforming it in specificity across mammalian datasets, though sometimes at the cost of sensitivity for low-abundance miRNAs (Lu et al. 2018).

ShortStack takes a genome-wide perspective, using Bowtie-based clustering to define small RNA loci and classifying them according to structural and abundance criteria. This enables annotation of both known and novel miRNAs alongside other small RNA classes, producing highly specific results with very few false positives. Its trade-off, however, is reduced sensitivity in certain animal datasets, particularly with shallow sequencing depth (Axtell 2013; Shahid and Axtell 2014). sRNAbench, meanwhile, provides a flexible profiling pipeline that integrates alignment, annotation, and isomiR analysis in a single workflow. Benchmarks show strong concordance with miRDeep2 in both known and novel miRNA discovery, while offering greater consistency across datasets, which makes it particularly valuable for comparative or large-scale studies (Aparicio-Puerta et al. 2019; Aparicio-Puerta et al. 2022). Together, these tools highlight a spectrum of trade-offs—miRDeep2 and miRge2.0 prioritizing discovery versus specificity, and ShortStack and sRNAbench emphasizing accuracy and reproducibility.

Normalization and differential expression analysis

After preprocessing, raw read counts must be normalized to account for differences in sequencing depth and sample composition. Common approaches include reads per million (RPM) and transcripts per million (TPM), which normalize counts relative to total sequencing depth or transcript abundance, respectively, enabling basic inter-sample comparisons (Elahimanesh and Najafi 2024). RPKM and its paired-end equivalent, FPKM, additionally adjust for transcript length, making them useful for gene-level expression comparisons across transcripts of different sizes (Zhao et al. 2020). However, these methods, originally designed for mRNA-seq, are less relevant for miRNAs because of their uniform short length, and they may not fully account for biases inherent in small RNA sequencing data.

For more rigorous analysis, statistical models that handle discrete count data have become the standard. DESeq2 and edgeR, though originally developed for mRNA-seq, have been widely applied in miRNA-seq analysis. Both methods normalize raw counts to account for library size and composition—DESeq2 through size factors and edgeR through the trimmed mean of M-values (TMM)—and model counts with a negative binomial distribution to test for differential expression (Love et al. 2014a; Chen et al. 2014). Importantly, their suitability for miRNA-seq has been demonstrated in studies like the one conducted by Hu et al. where they applied DESeq2 and edgeR to miRNA-seq data across 14 cancer types, showing that both tools produced consistent and biologically meaningful results, underscoring their robustness for miRNA expression studies (Hu et al. 2018). While DESeq2 is often preferred for larger datasets due to its variance stabilizing transformation (VST), which improves data homoscedasticity and downstream analyses such as clustering or PCA (Love et al. 2014b), edgeR is considered advantageous for smaller datasets or those with low-count miRNAs because of its flexible dispersion estimation (Chen et al. 2025). Thus, despite being originally tailored for mRNA-seq, both tools have proven effective and reliable for differential expression in miRNA-seq.

Beyond these general frameworks, dedicated miRNA analysis platforms have been developed to address the specific challenges of short read length, low counts, and isomiR variability. The QIAseq miRNA Differential Expression Toolbox is optimized for miRNA-seq data, incorporating generalized linear model (GLM)-based statistics and multifactor experimental designs tailored to miRNA features and annotations (Qiagen. 2025). Similarly, miRNA-TA integrates miRNA-specific normalization strategies with statistical testing, allowing improved sensitivity in detecting differential expression compared with generalized RNA-seq tools (Cer et al. 2014). Another resource, the UEA sRNA Workbench, provides an end-to-end pipeline for small RNAs including miRNA-seq, from quality control through normalization and differential expression. It incorporates miRNA-specific annotation and filtering modules, helping to minimize false positives and ensure biologically relevant results (Stocks et al. 2012). These specialized frameworks, while less commonly used than DESeq2 or edgeR, offer pipelines explicitly designed to accommodate miRNA-specific characteristics such as isomiR diversity and uniform transcript length.

Data enrichment

After identifying differentially expressed miRNAs, tools such as DIANA-mAP facilitate the next step by linking these miRNAs to their target genes and associated biological pathways (Alexiou et al. 2020). These tools leverage databases such as TargetScan, miRDB, and Miranda to predict potential mRNA targets based on sequence complementarity and evolutionary conservation of miRNA binding sites. These predictions are then validated using experimentally curated databases such as TarBase and miRTarBase, which contain reliable information on experimentally validated miRNA-target interactions. Once miRNA targets are identified, DIANA-miRPath provides pathway enrichment analysis, offering insights into how dysregulation of miRNAs affects cellular pathways involved in cancer progression. This tool is particularly useful for identifying key cancer-related pathways that may be regulated by miRNAs, including tumor suppressor and oncogenic pathways (Tastsoglou et al. 2023a). In addition, the analysis of the KEGG (Kyoto Encyclopaedia of Genes and Genomes) pathway, integrated within DIANA-miRPath and other platforms such as miRNet, helps map miRNA-regulated pathways to known signaling cascades involved in cancer biology. miRNet offers additional pathway and network-based analysis, helping researchers integrate miRNA data with gene expression and protein–protein interaction data to construct regulatory networks (Chang et al. 2020). Furthermore, platforms like ingenuity pathway analysis (IPA) allow visualization and exploration of the biological significance of miRNA-gene interactions by mapping them to known signaling pathways. By combining these tools, researchers can gain a comprehensive understanding of how differentially expressed miRNAs drive cancer biology, providing insight into potential therapeutic targets, diagnostic markers, and novel treatment strategies. Table 1 summarizes the pre-processing and data analysis tools.

Multi-parametric integration for microRNA analysis

Multi-Omics integration

The complexity of cancer requires an integrative approach where miRNA data is combined with other omics layers such as transcriptomics, proteomics, epigenomics, and metabolomics. Multi-omics integration provides a comprehensive understanding of miRNA function and its role in disease mechanisms. Since miRNAs regulate gene expression by targeting mRNAs, one of the most common approaches in multiomics integration is miRNA-mRNA interaction analysis. This is typically achieved by correlating differentially expressed miRNAs and mRNAs within the same dataset. Databases such as DIANA-TarBase, miRTarBase, and StarBase provide experimentally validated miRNA-mRNA interactions, filtering out false positives from purely computational predictions (Karagkouni et al. 2018; Ji Lee et al. 2015). More recently, tools that incorporate data from cross-linking immunoprecipitation methods (e.g., eclip, rip-seq) have been developed to strengthen evidence for direct mirna–mrna interactions, providing a more robust framework for integrative analyses (Nostrand et al. 2016; Vaňková Hausnerová et al. 2024). Network visualization tools like Cytoscape allow researchers to construct and analyze miRNA-mRNA interaction networks by integrating expression data, clustering miRNAs, and their targets based on functional similarities or pathway involvement (Wang et al. 2021). Additionally, weighted correlation network analysis (WGCNA) is widely used to identify coexpression networks between miRNAs and mRNAs, clustering them into modules that reveal miRNA-regulated gene networks implicated in cancer. To achieve these analyses, researchers often use RNA-seq data alongside miRNA-seq data, applying tools such as DESeq2 or edgeR to identify differentially expressed miRNAs and mRNAs and then correlating their expression levels to predict regulatory interactions (Ghafouri-Fard et al. 2023).

Epigenetic modifications, such as DNA methylation and histone modifications, also play a crucial role in regulating miRNA expression, either silencing or activating its transcription. Chromatin immunoprecipitation sequencing (ChIP-seq) is commonly used to identify histone modifications associated with miRNA promoter regions, integrating ChIP-seq data with miRNA expression profiles to determine whether histone modifications influence miRNA transcriptional activity. Yet, the resolution of ChIP-seq is limited for many miRNA promoters, which often reside within intragenic regions. To address this, ATAC-seq provides complementary information on chromatin accessibility, helping to determine whether specific miRNA loci are transcriptionally permissive (Buenrostro et al. 2015). Similarly, bisulfite sequencing enables DNA methylation analysis, revealing whether hypermethylation of CpG islands in miRNA promoter regions leads to their downregulation in cancer. Tools such as BS-Seeker2 and Bismark are used to map and quantify DNA methylation patterns, while publicly available ENCODE epigenetic

Table 1 Pre-processing and data analysis used in microRNA research

Tool name	Purpose/functionality	Input data types	Output	Method/algorithm	Availability	Comments	Reference
File conversion & quality control							
BCL2FASTQ	BCL to FASTQ conversion	BCL files	FASTQ files	File conversion	Illumina software	Standard Illumina tool for file conversion	Illumina. bcl2fastq. 2025)
Cutadapt	Adapter trimming	FASTQ files	Trimmed reads	Sequence trimming	Command-line	Highly customizable trimming	Martin 2011)
Trim Galore	Adapter trimming with QC	FASTQ files	Trimmed reads	Wrapper tool	Command-line	Wrapper around Cutadapt and FastQC	Krueger 2015)
miRNA Identification & Alignment							
miRDeep2	miRNA identification and quantification	FASTQ files	Known/novel miRNAs	Probabilistic model	Command-line	High sensitivity, prone to false positives	Friedländer et al. (2012)
miRge2.0	miRNA analysis with ML module	FASTQ files	miRNA counts	SVM-based	Command-line	Conservative approach, high specificity	Lu et al. (2018)
ShortStack	Small RNA annotation	FASTQ files	Small RNA loci	Clustering-based	Command-line	Genome-wide perspective, high specificity	Axtell (2013); Shahid and Axtell (2014)
sRNAbench	Small RNA profiling pipeline	FASTQ files	miRNA profiles	Integrated pipeline	Web/Command-line	Flexible pipeline with isomiR analysis	Aparicio-Puerta et al. (2019; Aparicio-Puerta et al. (2022)
Differential expression analysis							
DESeq2	Differential expression analysis	Count matrices	DE miRNAs	Negative binomial	R package	Preferred for larger datasets with VST	Niedziela et al. (2022)
edgeR	Differential expression analysis	Count matrices	DE miRNAs	Negative binomial	R package	Good for smaller datasets/low counts	Robinson et al. (2010)
QIAseq miRNA DE Toolbox	miRNA-specific DE analysis	miRNA counts	DE results	GLM-based	Commercial	miRNA-optimized statistical models	Qiagen (2025)
miRNA-TA	Temporal miRNA analysis	Time-series data	DE miRNAs	Quantile transformation	Tool	miRNA-specific normalization	Cer et al. (2014)
UEA sRNA Workbench	End-to-end small RNA pipeline	FASTQ files	Analysis results	Comprehensive pipeline	Software suite	Complete miRNA-seq workflow	Stocks et al. (2012)
Pathway & functional analysis							
DIANA-miRPath	Pathway enrichment analysis	miRNA lists	Pathway results	Enrichment analysis	Web tool	miRNA-pathway mapping	Papadopoulos et al. (2009)
DIANA-mAP	miRNA analysis pipeline	NGS data	miRNA quantification	Integrated pipeline	Web Tool	Raw NGS to quantification	Alexiou et al. (2020)
miRNet	Network-based miRNA analysis	miRNA data	Network visualizations	Network analysis	Web tool	Integrates multiple data types	Chang et al. (2020)
Ingenuity pathway analysis	Pathway visualization	miRNA-gene data	Pathway networks	Knowledge-based	Commercial	Comprehensive pathway analysis	UCSLibraries (n.d.)

Table 1 (continued)

Tool name	Purpose/functionality	Input data types	Output	Method/algorithm	Availability	Comments	Reference
Network analysis							
Cytoscape	Network visualization	Interaction data	Network graphs	Graph theory	Software	miRNA-mRNA network construction	Wang et al. (2021)
WGCNA	Co-expression network analysis	Expression matrices	Gene modules	Correlation-based	R package	Identifies co-expression networks	Ghafouri-Fard et al. (2023)

datasets facilitate integrative analyses (Guo et al. 2013; Krueger and Andrews 2011).

Multi-RNA integration

The study of multi-RNA frameworks like CircularRNA-miRNA-mRNA helps us gain a deeper understanding of the disease and thereby the underlying network. CircularRNAs (CircRNAs) function as competing endogenous RNAs (ceRNAs) by sequestering miRNAs through shared miRNA response elements (MREs), thereby modulating the expression of target mRNAs and creating complex regulatory networks. Recent studies have demonstrated the critical importance of circRNA-miRNA-mRNA interactome analysis in understanding cancer pathogenesis, particularly in ovarian cancer, where specific circRNAs such as hsa_circ_0001756, hsa_circ_0002711, and hsa_circ_0051240 act as oncogenic regulators by sponging tumor suppressor miRNAs hsa-miR-1288, hsa-miR-1229, and hsa-miR-637 respectively (Srivastava 2025). Furthermore, integrative studies in prostate cancer have constructed comprehensive mRNA–SNP–miRNA regulatory networks using multi-omics data and in silico predictions followed by TCGA-based validation to find robust diagnostic and prognostic biomarkers such as hsa-miR-21, hsa-miR-30b, and ITGB8. Other applications span ceRNA network analyses in solid tumors, where integrated lncRNA, miRNA, and mRNA datasets identify key drivers and therapeutic targets through network-based approaches and validation against patient outcomes (Li et al. 2022; Wang et al. 2022).

Several specialized bioinformatics tools have been developed to facilitate circRNA-miRNA-mRNA network construction and analysis. CircInteractome serves as a comprehensive web platform for mapping miRNA and RNA-binding protein sites on circRNAs, enabling researchers to predict ceRNA interactions and design experimental validation strategies (Dudekula et al. 2016). CRAFT (CircRNA Function prediction Tool) provides an integrated pipeline for predicting circRNA molecular interactions with miRNAs and RNA-binding proteins, while investigating complex regulatory networks (Dal Molin et al. 2022). These computational approaches enable comprehensive ceRNA network reconstruction and target prediction for the circRNA-miRNA-mRNA axis.

Machine learning approaches in analyzing microRNA data

Advancements in ML and AI are revolutionizing the analysis and interpretation of miRNA data in cancer research. Given the high dimensionality and complexity of multiomics datasets, traditional statistical approaches often fall short in capturing intricate patterns and nonlinear relationships between miRNAs and their molecular targets. AI-driven techniques provide a more robust framework for identifying differentially expressed miRNAs, predicting miRNA-mRNA interactions, and integrating multiomics data for biomarker discovery and disease classification.

A key application of ML in miRNA research is feature selection and classification, which enables the identification of miRNA signatures that can distinguish cancerous from noncancerous samples with high accuracy. Various supervised learning algorithms, such as SVMs, random forests, and deep neural networks, have been widely used for this purpose (Thomas and Sael 2017); He et al. 2018). SVMs are particularly effective for small to medium-sized miRNA datasets because they work by mapping input data into a higher-dimensional space and finding an optimal hyperplane that separates different classes. This makes them highly effective in handling nonlinearly separable data, a common challenge in biological datasets (Parveen et al. 2019; Mégret et al. 2022; Shakiba and Rueda 2013). In contrast, random forests, which are ensemble learning models based on decision trees, offer robustness against overfitting and improve classification performance by aggregating predictions from multiple decision trees. This method is particularly useful when dealing with noisy or high-dimensional miRNA expression data (Wei et al. 2022).

Deep neural networks (DNNs), particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have demonstrated exceptional performance in classifying tumor subtypes based on miRNA expression patterns. CNNs, which are typically used for image recognition, can also be applied to transcriptomic data by capturing spatial relationships in high-dimensional datasets, allowing them to detect complex miRNA expression patterns across cancer types (Gunavathi et al. 2021). Their ability to learn hierarchical features makes them particularly suitable for identifying subtle but significant variations in expression profiles (Bilal et al. 2017). On the other hand, RNNs are designed to process sequential data, making them ideal for analyzing time-series or longitudinal miRNA expression patterns (Li et al. 2023). Since miRNA regulation can be dynamic, RNNs can model dependencies in gene expression over time, providing deeper insights into miRNA-mediated regulatory mechanisms (Vineetha et al. 2013).

However, a crucial challenge in applying ml to rare cancer datasets is the small sample size (often $n < 100$), which increases the risk of overfitting. To address this, researchers commonly use cross-validation strategies such as leave-one-out validation, where each sample is iteratively held out as a test case while the rest form the training set, or leave-one-cancer-out validation, where an entire cancer type is left out to test generalizability across cancer contexts (Learn 2025a). These strategies provide more reliable estimates of model performance in scenarios with limited data and high dimensionality, ensuring that predictive models remain robust. Another key consideration is the interpretability of deep learning models. While CNNs and RNNs achieve high accuracy, they are often criticized as “black boxes.” Methods such as SHAP (SHapley Additive Explanations), which assigns each feature an importance score based on its contribution to the model’s prediction, and LIME (Local Interpretable Model-agnostic Explanations), which approximates the model locally with a simpler interpretable model to reveal feature influence, can be applied to deep models (SHAP 2025; Zafar and Khan 2021). These approaches highlight which miRNAs drive specific predictions, providing biological insight into model outputs and increasing their translational value for clinical applications.

miRdisNET and miTAR utilize advanced ML models to predict disease-associated miRNAs and identify miRNA-disease associations from large-scale datasets. miRdisNET employs a biological knowledge-based ML approach, incorporating known associations to improve prediction accuracy by applying a grouping, scoring, and modelling (G-S-M) framework (Jabeer et al. 2022). miTAR, on the other hand, integrates CNN and RNN to provide a comprehensive miRNA:target interaction network (Gu et al. 2021).

Unsupervised learning techniques help analyze miRNA expression data by identifying patterns within complex datasets. Hierarchical clustering groups similar miRNA profiles, aiding in patient subtype classification and detecting coexpressed miRNA clusters relevant to cancer. Principal component analysis (PCA) reduces the dimensionality of the data while preserving variance, allowing for better visualization and identification of key miRNAs (Sell et al. 2020). More advanced methods like t-distributed stochastic neighbour embedding (t-SNE) and UMAP map high-dimensional miRNA data into lower dimensions while preserving structure. t-SNE maps the data into lower dimensions by converting pairwise similarities into probabilities and optimizing local cluster retention through gradient descent, although it may struggle to preserve global structure (Wang et al. 2020; Tian and Tao 2020). UMAP, on the other hand, constructs a high-dimensional graph of data points and optimizes a lower-dimensional representation while maintaining both local and global relationships efficiently. This makes UMAP faster and better at capturing the overall structure of miRNA expression data (Yang et al. 2021). These techniques help stratify patients, discover biomarkers, and explore miRNA expression in oncology.

Beyond classification, ML and AI models play a crucial role in the prediction of miRNA-mRNA interactions. AI-based tools such as TargetNet, DIANA-microT, and mirDIP incorporate ML algorithms to refine target predictions by integrating expression data, sequence features, and experimental validation datasets (Min et al. 2022; Reczko et al. 2011; Tokar et al. 2018). Natural language processing (NLP)-based models are also being employed to mine vast amounts of scientific literature, automating the discovery of miRNA gene associations from published studies (Bhasuran et al. 2024). Figure 2 provides a simplified framework of the development roadmap of diagnostics and prognostic tools using AI/ML and DL.

Integrating ML with multiomics data further enhances our understanding of miRNA function in cancer. Multi-Omic Factor Analysis (MOFA) and iCluster are advanced frameworks that integrate miRNA-seq, RNA-seq, and proteomic data to identify co-regulated molecular signatures (Argelaguet et al. 2018; Tian and Wang 2021). Deep learning models, such as autoencoders, can be trained on large multiomics datasets to identify latent features that correlate with disease progression, treatment response, and survival outcomes. These models can be particularly useful for predicting drug sensitivity based on miRNA expression patterns, paving the way for precision oncology applications (Pratella et al. 2021).

Beyond diagnostics, computational models also hold promise for therapeutic discovery. By analyzing miRNA-regulated networks, ML and AI approaches can help identify novel drug targets and small molecules that modulate these pathways. For example, miR-7-5p has been shown to regulate Trefoil Factor 3 (TFF3) expression in intestinal epithelial cells, linking miRNA biology directly with therapeutic target discovery. These examples highlight how computational pipelines can bridge biomarker discovery

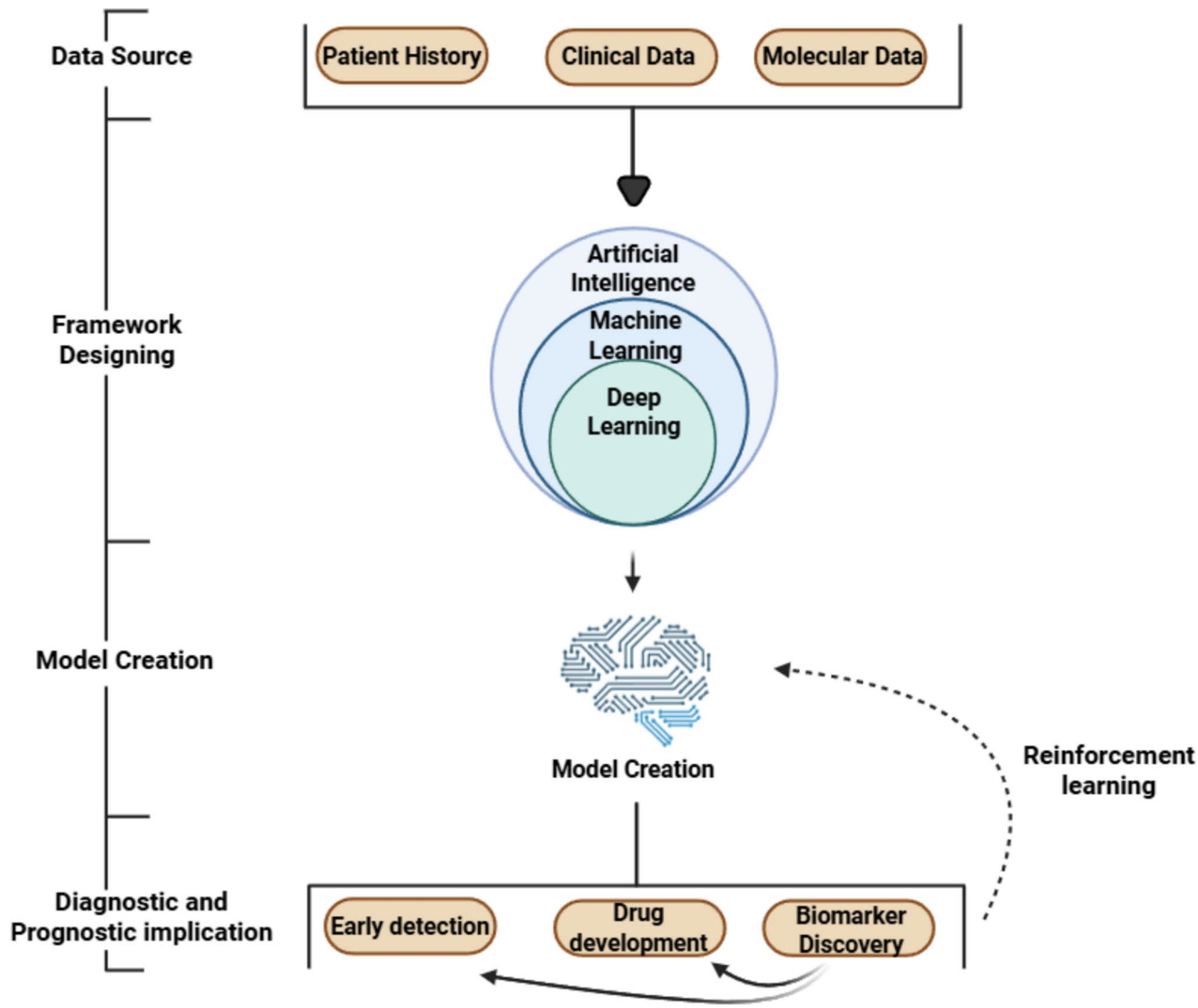

Fig. 2 Schematic framework of the development roadmap of diagnostic and prognostic tools using AI/ML and DL. Created in BioRender. Dey, M. (2025b)

with drug development, paving the way for precision oncology interventions (Zhang et al. 2019; Poh et al. 2019; Guo et al. 2017).

LLMs and generative AI are also emerging as powerful tools in miRNA research. LLMs trained in biomedical literature and large-scale sequencing data can assist in hypothesis generation, suggesting novel miRNA targets and regulatory pathways (Bhasuran et al. 2024). Additionally, generative adversarial networks (GAN) can simulate synthetic miRNA expression profiles, enhancing data sets for rare cancer subtypes where patient-derived samples are limited (Ahmed et al. 2021). Table 2 summarizes the AI and ML-based tools for microRNA identification and classification.

Clinical studies integrating bioinformatics, ML, and multi-omics approaches

Clinical studies increasingly demonstrate how bioinformatics and multi-omics approaches are transforming cancer research and treatment. In multiple myeloma, Srivastava showed how AI-driven integration of transcriptomic and epigenomic (ChIP-seq) datasets can reveal differentially expressed miRNA networks that not only illuminate disease pathogenesis but also point toward therapeutic routes (Srivastava 2023). A complementary approach by Wogy et al. used unsupervised learning and similarity network fusion (SNF) across multi-omics datasets to

Table 2 Machine learning and AI tools for miRNA analysis

Tool name	Task/application	Algorithm/model	Input Type	Output	Comments	Ref
Supervised learning algorithms						
Support vector machines (SVMs)	Classification/feature selection	Support vector machine	Expression matrices	Cancer classification	Effective for small-medium datasets, handles nonlinear data	Parveen et al. 2019; Mégret et al. 2022; Shakiba and Rueda 2013)
Random forests	Ensemble classification	Decision tree ensemble	Expression data	Classification results	Robust against overfitting, handles noisy data	Learn 2025b)
Unsupervised learning algorithms						
t-SNE	Non-linear dimensionality reduction	Stochastic neighbor embedding	High-dimensional data	2D/3D visualization	Preserves local structure, good clustering	learn, S. TSNE. 2025)
UMAP	Dimensionality reduction	Uniform manifold approximation	High-dimensional data	Low-dimensional embedding	Preserves local and global structure efficiently	McInnes et al. 1802)
Deep learning algorithm						
Convolutional neural networks (CNNs)	Tumor subtype classification	Convolutional networks	Expression patterns	Tumor classification	Captures spatial relationships, learns hierarchical features	Gunavathi et al. 2021)
Recurrent neural networks (RNNs)	Temporal pattern analysis	Recurrent networks	Time-series data	Temporal patterns	Models temporal dependencies in expression	Li et al. 2023)
TSK-type RNN Fuzzy Network	miRNA-mRNA interaction modeling	Fuzzy neural network	Expression data	Interaction networks	Specialized RNN variant for miRNA interactions	Vineetha et al. 2013)
Specialized miRNA ML Tools						
miRdisNET	Disease-miRNA association prediction	G-S-M framework	miRNA expression	Disease associations	Biological knowledge-based ML approach	Jabeer et al. 2022)
miTAR	miRNA-target interaction prediction	CNN + RNN hybrid	Sequence/expression data	Interaction networks	Integrates multiple deep learning architectures	Gu et al. 2021)
TargetNet	Functional target prediction	Deep neural networks	miRNA data	Target predictions	Deep learning for functional target prediction	Min et al. 2022)
mirDIP	Integrative target prediction	ML integration	Multiple databases	Integrated predictions	Integrative database with ML enhancement	Tokar et al. 2018)
MBSTar	ML-based target prediction	Multiple instance learning	miRNA data	Binding site predictions	Multiple instance learning for functional sites	Bandyopadhyay et al. 2015)
MiRTDL	Deep learning target prediction	Deep learning	Sequence features	Target predictions	Deep learning approach for target discovery	Shuang et al. 2016)
Generative AI & advanced models						
Generative adversarial networks (GANs)	Data augmentation	Generative adversarial training	Limited datasets	Synthetic miRNA profiles	Enhances datasets for rare cancer subtypes	Ahmed et al. 2021)
Large language models (LLMs)	Hypothesis generation	Transformer architectures	Biomedical literature	Novel insights	Trained on biomedical literature and sequencing data	Bhasuran et al. 2024)

Table 2 (continued)

Tool name	Task/application	Algorithm/model	Input Type	Output	Comments	Ref
Explainable AI methods						
SHAP (SHapley Additive Explanations)	Model interpretability	Game theory-based	ML model outputs	Feature importance scores	Assigns importance scores to individual features	SHAP 2025
LIME (local interpretable model-agnostic)	Local model interpretation	Local approximation	Complex model predictions	Local explanations	Provides interpretable local approximations	Zafar and Khan 2021

explore rational drug combinations, with their phase I trial reporting that atezolizumab monotherapy (Atezo) and its combinations with standard-of-care (SOC) regimens were both safe and exhibited early signs of efficacy (Wong et al. 2023).

These strategies have since expanded to a broader range of cancers. Pan-cancer studies employing integrated miRNA–mRNA–lncRNA interaction networks and machine learning have achieved remarkable accuracy—over 99% in distinguishing tumor tissue of origin across 14 different cancer types. Importantly, such models also identified prognostically significant miRNAs that are now entering clinical trials, exemplifying how computational pipelines can move seamlessly from discovery to translational application (Lawarde et al. 2025). In parallel, diagnostic innovation has accelerated through the adoption of explainable AI frameworks that combine ensemble methods such as decision trees, neural networks, and SVMs (Chen et al. 2019). Leveraging circulating miRNA profiles, these approaches have produced clinically validated, non-invasive diagnostic tools, with the i-Biomarker CaDx test in breast cancer standing out for achieving 100% diagnostic accuracy while offering mechanistic insight into miRNA-driven tumor biology (Floares et al. 2023).

Together, these studies underscore the growing clinical relevance of bioinformatics-powered, multi-omic strategies, ranging from therapy design to diagnostic precision, signaling a future where miRNA-centred analytics play a central role in precision oncology.

Challenges and future perspectives

Despite rapid advancements in bioinformatics tools for miRNA research, several challenges remain to achieve highly accurate and clinically relevant analyses. One of the primary challenges is the variability of data arising from differences in sequencing platforms, library preparation methods, and computational pipelines. Variability in normalization techniques and data preprocessing steps can significantly impact differential expression analysis, making standardization essential for reproducibility (Gonzalo-Calvo et al. 2022). Furthermore, while databases like miRTarBase, TargetScan, and DIANA provide extensive miRNA-mRNA interaction data, inconsistencies between different prediction algorithms and the limited availability of experimentally validated interactions pose challenges in interpreting biological significance (Tabas-Madrid, et al. 2014; Kariuki et al. 2023).

The integration of multi-omics data presents another major hurdle. Although combining miRNA expression with transcriptomic, proteomic, and epigenomic datasets offers a more holistic view of cancer biology, the sheer complexity of regulatory interactions demands advanced computational models capable of handling high-dimensional data (Vahabi and Michailidis 2022; Zheng et al. 2024; Gumbs et al. 2023). Network-based approaches such as WGCNA and Cytoscape have made strides in visualizing interactions, but the challenge lies in accurately modelling dynamic regulatory mechanisms and feedback loops in cancer progression (Vahabi and Michailidis 2022).

ML and AI hold promise for miRNA research, but face challenges such as limited dataset size, overfitting, and data imbalance. Traditional ML models, including SVMs and random forests, struggle with high-dimensional data, which affects computational efficiency and accuracy (Saçar and Allmer 2013). Deep learning methods, such as convolutional and recurrent neural networks, excel in miRNA-based cancer classification and clustering, but suffer from interpretability issues, limiting their clinical adoption. The "black-box" nature of these models remains a major barrier to their integration into precision medicine (Shommo et al. 2024; Jiang et al. 2023). Addressing this requires the incorporation of explainable AI (XAI) frameworks, hybrid models that combine mechanistic biology with ML predictions, and transparent model reporting to clinicians.

Similar problems arise with whole-genome sequencing (WGS) as well. It exemplifies both the promise and current limitations of large-scale genomic profiling in oncology. Despite its ability to capture the full mutational and regulatory landscape, WGS is not yet widely adopted in clinical practice due to high costs, massive data storage and analysis demands, and the lack of standardized pipelines for variant interpretation across laboratories (Alagarswamyl et al. 2024). Regulatory considerations also represent a critical bottleneck. Clinical adoption of miRNA-based biomarkers demands rigorous validation across independent cohorts, multi-center trials, and alignment with regulatory frameworks such as FDA and EMA biomarker qualification programs. Furthermore, ensuring compliance with data privacy regulations (e.g., HIPAA, GDPR) and establishing standardized benchmarks for model performance will be essential for clinical deployment. Cost-effectiveness also poses a challenge, as widespread adoption of miRNA diagnostics requires demonstrating that these tests improve patient outcomes in a manner proportionate to their financial burden on healthcare systems.

Several strategies are being explored to overcome these barriers. Federated learning approaches allow models to be trained across multiple institutions without sharing raw patient data, addressing both data scarcity and privacy concerns (Yurdem et al. 2024). Public–private partnerships can accelerate biomarker validation by pooling expertise and resources from academia, industry, and healthcare systems. Adaptive clinical trial designs provide another solution, enabling dynamic incorporation of novel miRNA biomarkers or therapeutics as evidence emerges, thereby increasing trial efficiency and translational potential.

Ultimately, the future of miRNA bioinformatics lies in building robust, interpretable, and regulation-ready pipelines. Strategies such as the creation of global consortia to harmonize data standards, incorporation of explainable AI to foster clinical trust, and proactive collaboration with regulatory agencies can accelerate translation into practice. By addressing these barriers, miRNA-based diagnostics and therapeutics can progress from research settings into reliable tools for early detection, prognosis, and personalized oncology.

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